

- QUESTION 1 If $z = 1 + i\sqrt{3}$ express z in the following form 2
- $$z = r(\cos\theta + i\sin\theta)$$
- and hence plot on the Argand diagram:
- i) $\bar{z}$ 1
- ii) $\frac{12}{z}$ 1
- iii) z^2 1
- iv) $\sqrt{z}$ 1
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- QUESTION 2 Solve for x and y : 3
- $$(x + iy)(2 + 3i) = 18i - 1$$
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- QUESTION 3 Sketch on the Argand diagram where 3
- $$|z| < 4 \quad \text{and} \quad \frac{-\pi}{3} \leq \arg z \leq \frac{\pi}{4} \quad \text{both hold.}$$
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- QUESTION 4 Prove that $(6 + i\sqrt{108})^{12}$ is real. 2
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- QUESTION 5 Find the modulus of the product of the roots of the equation: 2
- $$(2 + 3i)z^2 + (5 - 2i)z + (18 + i) = 0$$
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- QUESTION 6 Find $\sqrt{-8 - 6i}$ and hence solve the equation: 4
- $$z^2 - (5 - i)z + 8 - i = 0$$

- QUESTION 7 P, Q and R represent the complex numbers 4
- z_1, z_2 and z_3 respectively on the Argand diagram.
- Given that $z_1 - z_2 = i(z_2 - z_3)$ show this information on a diagram and hence fully describe ΔPQR .
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- QUESTION 8 The point P on the Argand diagram represents the complex 3
- number z where z satisfies the equation:
- $|z - 2i| = \text{Im}(z + 2i)$
- Find the Cartesian equation of the locus of P and give a geometric description of the locus of P.
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- QUESTION 9 By applying De Moivre's Theorem, and by also expanding 4
- $(\cos\theta + i\sin\theta)^4$ express $\tan 4\theta$ as a polynomial in terms of $\tan\theta$.
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- QUESTION 10 i) If ω is a complex solution of the equation $z^3 = 1$ show that ω^2 2
- is also a solution of the equation and that:
- $1 + \omega + \omega^2 = 0$
- (ii) Given that ω is also a root of $z^3 + bz^2 + cz - 12 = 0$, 2
- find the values of b and c given they are both real.

END OF EXAMINATION

Question 1

$$z = 1 + i\sqrt{3}$$

$$= 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$$

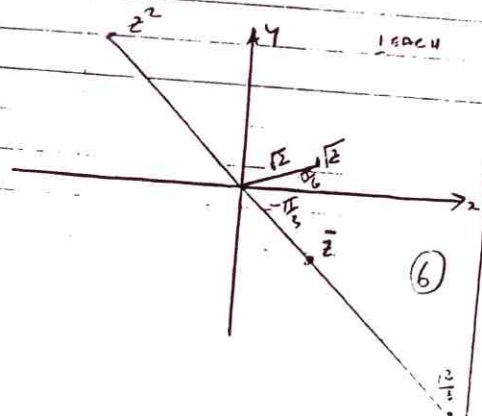
$$= 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$\bar{z} = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$\frac{12}{2} = 6\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$z^2 = 4\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

$$\sqrt{z} = \sqrt{2}\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$



(6)

Question 2

$$(x+iy)(2+3i) = 18i - 1$$

$$(2x-3y) + i(2y+3x) = 18i - 1$$

$$\therefore 2x-3y = -1 \quad (1)$$

$$2y+3x = 18 \quad (2)$$

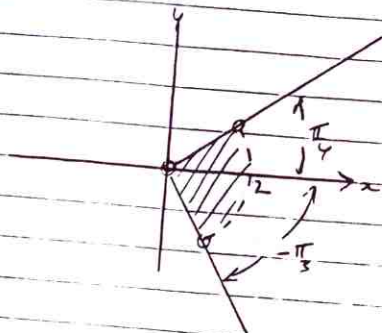
$$\times 2 \Rightarrow 4x-6y = -2 \quad (3)$$

$$\times 3 \Rightarrow 6y+9x = 54 \quad (4)$$

$$+ (4) \Rightarrow 13x = 52 \quad (3)$$

$$x = 4$$

Question 3



$\angle = \frac{\pi}{4}$

$\frac{\pi}{4}$

1. ...

Question 4

$$(6+i\sqrt{108})^{12} = (6+6\sqrt{3}i)^{12}$$

$$= 12^{12} \left(\frac{6+6\sqrt{3}i}{12}\right)^{12}$$

$$= 12^{12} \left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{12}$$

$$= 12^{12} \left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{12}$$

$$= 12^{12} \left(\cos 4\pi + i\sin 4\pi\right)$$

$$= 12^{12} (\cos 0 + i\sin 0)$$

$$= 12^{12} \quad -1$$

WHICH IS REAL

(2)

Question 5

$$(2+3i)z^2 + (5-2i)z + (18+i) = 0$$

$$\Delta B = \frac{c}{a} = \frac{18+i}{2+3i} \times \frac{2-3i}{2-3i} = -1$$

$$= (36+3) + i(2-54)$$

$$= 39 - 52i$$

$$= 39 - 52i$$

$$= 39 - 52i$$

$$= 39 - 52i$$

$$= 39 - 52i$$

$$|AB| = \sqrt{3^2 + 4^2} = 5$$

Q7 (cont) $\triangle PQR$ IS AN ISOSCELES
RIGHT ANGLED TRIANGLE
WITH $R \angle$ AT $Q(z_2)$.

Question 6

$$\text{Let } z = x+iy$$

$$\therefore x+iy = \sqrt{-8-6i}$$

$$\therefore x^2-y^2+2xyi = -8-6i$$

$$\therefore x^2-y^2 = -8$$

$$2xy = -6$$

$$\text{mult } y = \frac{-3}{x}$$

$$x^2 - \frac{9}{x^2} = -8$$

$$\therefore x^4 + 8x^2 - 9 = 0$$

$$(x^2+9)(x^2-1) = 0$$

$$\therefore x = \pm 1 \quad y = \pm 3$$

$$\therefore z = 1-3i$$

$$3^2 - (5-i)^2 - 4(5-i) = 0$$

$$3 = \frac{5-i \pm \sqrt{(5-i)^2 - 4(5-i)}}{2}$$

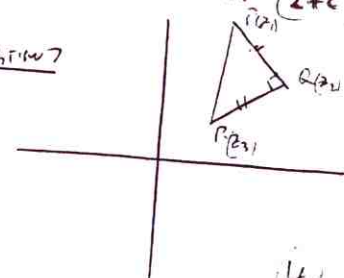
$$= \frac{5-i \pm \sqrt{25-10i-1-32+4i}}{2}$$

$$= \frac{5-i \pm \sqrt{-8-6i}}{2}$$

$$= \frac{5-i \pm 1-3i}{2} \text{ or } \frac{5-i-1+3i}{2}$$

$$= 3-2i \text{ or } (2+i)$$

Question 7



(4)

Question 8.

$$|z - zi| = \ln(z + zi)$$

Let $z = x + iy$

$$|(x + iy - zi)| = \ln(x + iy + zi)$$

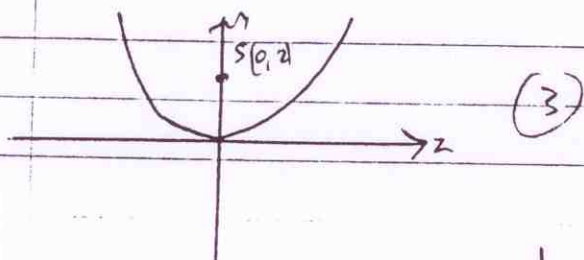
$$|x + i(y - z)| = y + z \quad \perp$$

$$\therefore \sqrt{x^2 + (y - z)^2} = y + z$$

$$x^2 + y^2 - 4y + 4 = y^2 + 4y + 4$$

$$x^2 = 8y \quad \perp$$

$$\therefore x^2 = 4 \cdot 2y.$$



A Parabola Vertex O and focus (0,2)

Question 9

$$(\cos 4\theta + i \sin 4\theta) = (\cos \theta + i \sin \theta)^4 \text{ Let } z = \cos \theta, i = \sin \theta$$

$$\cos 4\theta + i \sin 4\theta = C^4 + 4C^3Di + 6C^2D^2i^2 + 4CD^3i^3 + D^4i^4 \quad \perp$$

$$= C^4 + 4C^3Di - 6C^2D^2 - 4CD^3i + D^4 \quad \perp$$

$$\cos 4\theta = C^4 - 6C^2D^2 + 1 \quad \sin 4\theta = 4C^3D - 4CD^3 \quad \perp$$

$$\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta} = \frac{4C^3D - 4CD^3}{C^4 - 6C^2D^2 + 1} \quad \perp$$

$\therefore$ top + bottom by C^4

$$\therefore \tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta} = \frac{4\frac{D^2}{C^2} - 4\frac{D^4}{C^4}}{1 - 6\frac{D^2}{C^2} + \frac{D^4}{C^4}} \quad (4)$$

$$= \frac{4\sin^2\theta - 4\sin^4\theta}{1 - 6\sin^2\theta + \sin^4\theta} \quad \perp$$

Question 10

1) $z^3 = 1$

$$\therefore (z-1)(z^2+z+1)$$

$$(3-1)(3^2+3+1)=0$$

if w is a complex root & then eqn

$$w \neq 1 \quad \perp$$

$$\therefore w \text{ must satisfy } z^2 + z + 1 = 0$$

$$\therefore w^2 + w + 1 = 0 \quad \perp$$

OR if w is a root

$$\text{then } w^3 = 1$$

$$\text{but } (w^2)^3 = (w^3)^2 = 1 \quad \perp$$

$$\therefore w^2 \text{ is also a root}$$

$$\text{now } z = 1$$

$$\therefore 3 \text{ roots are } 1, w, w^2$$

$$\text{now } \Sigma d \text{ for } z^3 - 1 = 0$$

$$\Sigma d = -\frac{b}{a} = 0 \quad \perp$$

$$\therefore 1 + w + w^2 = 0$$

11) if w is a solution so is w^2

$$\text{now } \alpha \beta \gamma = -\frac{d}{a} = 12$$

$$w - w^2 \cdot \gamma = 12$$

$$\gamma = 12$$

$$\Sigma d = 12 + w + w^2 = -b$$

$$\therefore 11 + (1 + w + w^2) = -b \quad \perp$$

$$\therefore b = -11 \quad \neq$$

$$\alpha \beta + \beta \gamma + \gamma \alpha = w^3 + w\gamma + w^2\gamma = c$$

$$= 1 + \gamma(w + w^2)$$

$$= 1 + 12(w + w^2) = c$$

$$= 1 - 11(1 + w + w^2) - 12 = c$$

$$\therefore c = -11 \quad \neq \quad \perp$$